



Current State:

The current military and DoD pilot training process is costly and time-consuming. As the demand for pilots outpaces the available supply, there is a need for a more efficient way to train pilots. UAS technology has the potential to reduce the training requirements for pilots and operators, making it easier and faster to fulfill critical mission sets.

-Pilot Training & Mission Planning

Military and DOD pilots currently require 500-800 hours of flight training, costing \$5-12 million per pilot. The number of qualified pilots is not keeping up with the demand, leading to a deficit of 1600+ pilots in the USAF alone. The gap for defense, cargo, and commercial applications is greater, as experienced pilots are a rare commodity.

There are various defense specific training programs, and in order to progress through these programs, pilots complete a pre-flight school in 6 - 8 weeks. Pre-flight school helps trainees build basic flight knowledge, and from there they move on to primary school. Primary schools help pilots to gain comfort flying various entry level aircraft, such as the T6. From there, pilots move on to intermediate flight training, where they learn about systems knowledge for both instruments and ratings systems, accruing 40 - 80 hour flight time. Once they pass intermediate training, they move on to advanced flight school, where they learn tactics for flying in formation and at low levels, as well as landing in different conditions, including aircraft carriers for the Navy.

Military pilots undergo extensive training and must complete many hours of flight time before they are assigned to a particular aircraft model. Once they are assigned to a unit, they continue to accrue hours while participating in mission planning, training, and actual live flight operations. The complexity of a rehearsal flight or live mission can determine the amount of preparation and after-action review that is necessary, similar to the below table. The pre-mission planning stage for a more involved flight can include multiple flights that need to be coordinated in order to achieve the desired outcome. Actual flight can be more complicated and time consuming, as well as require a longer post-mission debrief.

Flight Complexity	Pre-Mission Plan	Live Flight Operation	Post Mission Debrief
Training - Rehearsal	30 - 60 min	15 - 45 min	15 - 30 min
Shorter – Low Complexity	1 - 2 hr	0.5 - 1.0 hr	1 - 2 hours
Longer – Higher Complexity	3 - 4 hr	45 - 90 min	4 - 8 hours

-Unmanned Aerial Systems

The use of unmanned aerial systems, or UAS, has the potential to reduce the training requirements for pilots and aircraft operators. UAS capabilities, along with various attachment sensor/payloads, provide commanders with intelligence, surveillance, and reconnaissance (ISR) capabilities. This allows for commanders to make decisions more quickly and effectively through a variety of aircraft sizes (Small Group 1, Medium Group 3, and Large Group 5). Additionally, UAS systems are becoming more capable, serving human operators with more relevant decision making information. This frees up cognitive load for both analysts and commanders.



Emerging Technologies:

With the advances in XR technologies, extended reality pilot training platforms are becoming more realistic, which can help pilots fly multiple aircraft in a coordinated manner. Additionally, AI agents are becoming increasingly capable and can be used to help pilots improve their flying skills. Autonomous and teamed systems are also becoming increasingly important and can be used for a variety of purposes, including intelligence, surveillance, and reconnaissance (ISR), as well as combat operations and advanced flight behaviors.

-Extended Reality – XR

The use of augmented and extended reality is revolutionizing pilot training platforms, making them more realistic every day. For example, [Red 6](#)'s augmented reality platform can now be used outdoors, which is a critical development for air combat training. This technology allows for a new paradigm in training by bringing the full potential of Live, Virtual and Constructive training to the real world. As XR capabilities advance, simulation will be able to more accurately mimic real-world training environments, which could eventually lead to video game-like capabilities that can be used to establish mission rehearsal capabilities for multiple small-large aircrafts, operated in the same LVC environment, flown by pilots across the world.

-AI Agents & Unmanned-Manned Teaming

The use of artificially intelligent agents is becoming increasingly common in complex tasks, such as flying aircraft. These RL (reinforcement learning) capabilities are demonstrated in complex AI demonstrations such as [Alpha Go](#), [Alpha Star](#), [Alpha Zero](#), and Darpa [AlphaDogfight Trials](#). As these systems become more advanced, pilots can use them to improve and sharpen their flying skills by practicing against super-human agents. Aircraft systems are also being enhanced with XR capabilities, which will merge training and aircraft systems to provide unprecedented real-world flight training. This will speed up the pilot training process and maximize lifetime flight mastery for a given pilot.

The advantages of flying multiple aircraft are manifold. First and foremost, it allows for increased coverage and intelligence-gathering potential. By having multiple aircraft flying a mission, the area being monitored can be more comprehensively scouted and surveilled. Additionally, it allows for more accurate and timely intelligence analysis and dissemination, as the data collected by each aircraft can be more easily processed and understood in the context of the overall mission. Finally, it also frees up cognitive load for operators, who can focus on more important tasks with the knowledge that the UAS systems are autonomously flying and gathering intelligence.

As highlighted with AI Agent training and teaming, autonomous systems are becoming increasingly important in the military domain. They can be used for a variety of purposes, including intelligence, surveillance, and reconnaissance (ISR), as well as combat operations. However, the use of autonomous systems raises a number of issues, including the potential for accidents and the risk of cyber-attacks. One of the biggest challenges facing the development of autonomous systems is the need to create systems that are able to operate safely and effectively in complex environments. This requires the development of sophisticated algorithms and decision-making systems that can handle the large amount of data that is often encountered in military operations. Another challenge facing the development of autonomous systems is the need to ensure that these systems can be effectively operated by human beings. This requires the development of user



interfaces and training programs that can allow operators to effectively control and monitor the autonomous systems. The development of autonomous systems is a complex and challenging process, but it is clear that they have the potential to play a significant role in the future of the military.

- Advanced Flight Behaviors

There are a number of ways in which RL methods can be used to develop flight behaviors for specific mission sets. For example, low altitude flying or other low terrain flights can be developed for specific mission sets. Additionally, RL methods could be used to maximize survivability, when threats arise, to temporarily execute evasion maneuvers that thwart an oncoming attack by an advancing weapon, missile, or aircraft systems.

Technology Opportunities:

VR and AR technologies (aka: XR – Extended Reality) can improve training simulations for pilots, by providing a realistic environment in which they can train. These technologies can improve the speed to qualifying flight proficiency and help reduce the cost of pilot training. Additionally, the use of VR and AR can help measure a team's effectiveness when executing a simulated mission. XR based training technologies can also help improve their proficiency in executing specific mission sets, and reduce the risk of friendly aircraft being put at risk during live training exercises.

- Pilot Training and Mastery Toolsets

The increasing affordability of VR and AR technology can improve training simulations for pilots, by providing a realistic environment in which they can train (see [Cesium](#) and [Gau Gan](#)). These technologies can improve the speed to qualifying flight proficiency and help reduce the cost of pilot training. Additionally, the use of VR and AR can also help measure a team's effectiveness when executing a simulated mission. XR based training technologies can also help improve their proficiency in executing specific mission sets, and reduce the risk of friendly aircraft being put at risk during live training exercises. The use of proficiency scores along with metric-based training can also help identify pilot gaps that need to be addressed in order to improve their performance on certain mission sets.

- Decision Making Speed

With the advent of teaming and autonomous aircraft capabilities, it is possible to improve situational awareness to relay pertinent information in near real-time to air, ground, and joint force commanders. AI generated insights would help commanders understand real-time changes and assess risks and threats over an area of operation. Pilots and their aircraft systems can act be a first pass filter to send, approve, and escalate relevant information, removing them from traditional control based paradigms of flying and controlling sensor/payload systems. This allows analysts to focus attention on task management planning around actionable intelligence. Speed to decisions and the delivery of effects is critical for digitized forces to have a competitive edge.

- Multi-System Flexibility

As unmanned aerial systems (UAS) become more advanced and trustworthy, their use in collaborative missions with manned aircraft and ground troops will increase. By allowing UAS to team with other UAS and manned aircraft, commanders will have more flexibility in adapting to changing mission needs and have an



asymmetric advantage against potential enemies. As unmanned aerial systems (UAS) become more complex, commanders will need to be able to download new capabilities and mission sets through cyber-secure updates to their aircraft in order to take advantage of their full potential. The proposed concept of operations for UAS modularity and machine learning provides commanders with the ability to optimize their missions by changing the type of UAS deployed, as well as the machine learning models used. This allows for a more tailored mission to the specific area of operation. The flexibility of maritime-based ISR missions provides commanders with the ability to optimize their aircraft and flight behaviors as needed for different types of missions. In particular, commanders can transition between maritime-based missions to city-urban ISR mission in order to best meet the needs of their changing objectives. This flexibility is important for commanders because it allows them to adapt their strategies as needed in order to achieve their goals.

The advancement of technology has made it easier to marry video gaming methods with real-world applications. As a result, it is now possible to have 6 small, medium, aircraft, and large aircraft in the same simulation and ground personnel interact to generate more real-time situational awareness. In addition, hardware in the loop toolsets allow one to plug and play hardware into simulation to mimic sensory input, in order to test out compute and other failure modes. This saves valuable resources during the verification and validation testing cycles. AI toolsets and other real-time rendering engines also improve the ability to perform teaming missions in simulation, while also rapidly prototyping how best to set up a system to provide real-time situational awareness on a semi-autonomous schedule. When these teamed aircraft provide us insights, synthesizing a team mate, to make more effective decisions with human check points, our force's safety are protected. This highlights the need to simulate and test these capabilities as the technology approaches cutting edge applications.

-AI Trust Building & Future Considerations

The use of artificial intelligence (AI) systems in advanced mission-sets presents a unique challenge for human operators. Before entrusting AI systems with complex tasks, humans must build familiarity and integration so the system can effectively fulfill the role of a synthesized team mate. In addition to training, simulation is critical for accelerating human factors-based interfaces. The data associated with these simulations must be consumable for human decision making, and also redundant in case one of these flexible semi-autonomous systems is neutralized. Once simulation capabilities incorporate AI agents designed for human interaction, design teams and future customers can rapidly prototype interfaces that are maximized to unmanned and manned teaming of various UAS systems.

The considerations for safety and lifetime mastery must both accelerate pilot proficiency through tools that enhance performance for a given pilot. For example, one pilot may find value in a certain UX interface, based off their proficiency in executing Task A with AI guided capabilities and recommendations. That said, another pilot may already have mastered Task A, and require help in another area: Task B. Therefore customization to some level may be required, as pilots learn how to interact with the systems that are enabling a higher proficiency at a task by task level.



Future State:

The use of AI-unmanned systems offers potential capabilities for pre-mission planning, live flight operation, and post-mission debrief. Recent technology advancements make it possible to offer solutions for all three categories prior to operating an AI capable unmanned-manned system. For example, mission rehearsal is possible prior to entering an aircraft. Additionally, augmented reality capabilities can be used to simulate adversarial dogfights. Traditionally, full-motion-video capabilities are used to raise situational awareness to decision makers. However, with good enough systems and associated detection models, less data/info packets are required for transmission to commanders and other relevant officers in the theater.

-Pilot Training & Proficiency

Before executing teamed Unmanned Aircraft System (UAS) missions with medium and large aircrafts, operator and commander proficiency must be trained and simulated in order to build the equivalent of a flight of experienced pilots commanding various aircraft systems. Similar to pilot training, the tools, simulations, and flight training must build pilot capabilities as they learn to master AI capable systems, while the system provides them with more and more capabilities, focused on reducing cognitive load and improving overall mission capabilities. Commanders also must understand the resource and capabilities at their disposal to push the boundaries and comfort at an individual level; as proficiency will vary by each individual utilizing the teamed aircraft controlled by one or many operators in charge of each teamed unit.

-Mission Planning & Post Mission Debriefs

The use of AI unmanned-manned systems offers potential capabilities for pre-mission planning, live flight operation, and post-mission debrief. Recent technology advancements make it possible to offer solutions for all three categories prior to operating an AI capable unmanned-manned system. For example, mission rehearsal is possible prior to entering an aircraft. Additionally, augmented reality capabilities can be used to simulate adversarial dogfights. Traditionally, full-motion-video capabilities are used to raise situational awareness to decision makers. However, with good enough systems and associated detection models, less data/info packets are required for transmission to commanders and other relevant officers in the AO.

The use of mobile sensors from various aircraft can aid in the creation of a more realistic battlefield environment and improve the ability of pilots and commanders to make informed decisions. However, the fidelity of these video feeds can be limited, hampering the ability of pilots and commanders to make informed decisions. In using **Live2Sim** toolsets, less digital data transfer of video would be necessary, and a more realistic view of the battlefield could be established near decision makers, improving situational awareness and decision making. Additionally, pilots could train in simulated environments, and try out various scenarios to build proficiency in mission rehearsal exercises. Post mission debriefs could also be improved, leading to the potential practice of additional training exercises built around a prior post mission debrief.



Conclusion:

Military and Department of Defense (DoD) pilots currently undergo extensive and costly training in order to be able to fly aircraft. However, advances in XR technologies (VR and AR) have the potential to reduce the training requirements for pilots, making it easier and faster to fill critical mission sets. Additionally, the use of AI-unmanned systems offers potential capabilities for pre-mission planning, live flight operation, and post-mission debrief. These technologies have the potential to improve the speed and efficiency of the pilot training process.

The use of technology today helps build pilot and UAS operators faster and also improves their proficiency and performance. However, as companies push the cutting edge, simulation and other real-time training becomes more critical to not only rapidly prototype capabilities, but also ensure humans and their systems work together as a team. This raises situational awareness and understanding, which in turn improves decisions.



I'm looking forward to completing Harvard's Business Analytics Program **#HBAP**. When embarking on the journey, I remember the first class taught by Karim Lakhani Digital Strategy and Innovation, and writing a paper around AI computer vision capabilities specific to Thermal Imaging. The below article was written the help of GPT3, and is of my own opinion to highlight the importance of simulation and training; building on some concepts from ITSEC 2021. I'd love to hear everyone's thoughts



Summary of Simulation Thoughts from ITSEC: Stick Notes

Metrics

- Speed
- Efficiency
- Decision
- Speed to Site
- Speed to Effect

Teaming

- Policy
- SATCOM

Per CRAFT

- Variants

Payload

- Modularity
- SWAP
- Ease

Integration

- Group 3
- Group 5
- UGV/etc

Decision

- Packets
- Approvals

Multi-Domain

- Info
- Execution

Theme 1

CU-T

TRUST

Building

- Simulation
- Teamed + Individual Capability Assessment
- Shareware Evidence
- Practice → Live (sim)
- Actual A2 Flight (Human Control of all assets)

Lifetime Value of Pilot

5-12 m

- Flight School
- Intermediate
- Advanced
- AIRCRAFT Assignment
- SWARMION
- TRADU → Airbase
- Live Mission

T2

Safety + Proof of UUV (Validation)

- Take off/Landing
- Release Conditions
- Hazard Identification
- Situational Awareness
- Route Guidance - Target
- Inclement Weather

T3

Lifetime Mastery

Level 1-5

- Manual Control of Operator
- Manual → AI/Control
- Semi-Manual CU-T Life
- CU-T Heavy (Usage)
- Mission Evolution

T4

Training Proficiency

- Art Metrics
- Task Skill Assessment

T5

Task Guiding

- Guiding of Proficiency
- MMK - Dayfighting
- Measure Command
- Believing
- Measure Post-class
- Leave & Cognitive Load

T6

Training to

- Pre Mission
- Flight Address
- Post Mission

T7

Custom UX

CU-T

- Human Optimized for Needs
- Menus Sized / Posed
- Options of Toolsets
- Adaptive per Mission Needs
- A2 learns Human

Avg Flight

Phase	1-2	3-4	5-6	7-8
Pre-Mission	1-2	3-4	5-6	7-8
Flight	1-2	3-4	5-6	7-8
Post-Mission	1-2	3-4	5-6	7-8